

Postpartum Hemorrhage 05: Bakri, Embolization, Surgery, and Hysterectomy

Companion reading handout for simultaneous listening.

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Bakri, embolization, surgery, and hysterectomy.

The last episode taught the four doors.

Tone, trauma, tissue, and thrombin.

Now we talk about what happens when the first measures are not enough

This is the escalation episode.

The tone of this lecture is deliberate.

Escalation is not panic.

Escalation is disciplined movement toward source control.

In postpartum hemorrhage, the danger is not only bleeding. The danger is late bleeding control, because delay changes the patient.

At the beginning, the patient may be warm, compensated, clotting, and salvageable.

Later, she may be cold, acidotic, coagulopathic, and unstable.

The same operation becomes harder.

The same suture holds less well.

The same bleeding surface clots less well.

The same decision costs more.

So here is the frame for this episode:

The ladder tells you the options.

The clock tells you whether you are allowed to use them.

Bakri balloon.

Embolization.

Laparotomy.

Compression sutures.

Uterine artery ligation.

Internal iliac ligation.

Packing.

Hysterectomy.

Those are rungs on a ladder.

But the patient's physiology is the clock.

If she is stable, you may have time for a fertility-preserving option.

If she is deteriorating, the correct answer may be definitive source control now.

Let us begin with intrauterine tamponade

Tamponade means pressure.

The Bakri balloon presses from inside.

It fills the uterine cavity with sterile saline so the balloon compresses the bleeding endometrial surface.

That mechanism is why it is useful in atony-related bleeding after uterotonics, when tears are not the primary source.

The Bakri is not magic.

It is internal pressure.

And because it is internal pressure, it only makes sense when the bleeding surface can be compressed from inside.

Before placing a Bakri, inspect the cervix and vagina.

Rule out tears as the main source.

A balloon inside the uterus will not repair a cervical laceration.

It will not fix arterial bleeding.

It will not fix uterine rupture.

It will not solve a major cavity-distorting anomaly.

The local protocol lists important contraindications and cautions: suspected arterial bleeding, suspected uterine rupture, cervical malignancy, some uterine anomalies such as a large cavity-distorting fibroid, purulent vaginal or cervical discharge or clear chorioamnionitis, and allergy to the balloon material.

The protocol also notes that guidance after cesarean is not completely clear and requires judgment.

The insertion principles are practical.

Empty the bladder with a Foley catheter.

Give adequate pain relief.

Disinfect the cervix and vagina.

Inspect the vagina and cervix.

Insert the balloon into the uterine cavity above the internal os, manually or with forceps.

Use ultrasound guidance when possible to confirm position.

Do not use excessive force.

If resistance persists, readjust or abandon the procedure.

The balloon is filled with sterile normal saline until resistance is encountered and bleeding stops or weakens.

The usual local volume is 250 to 300 milliliters.

The maximum is 500 milliliters.

Never inflate it with air or carbon dioxide because of the risk of air embolism.

Local protocol recommends Augmentin 1 gram intravenously every 8 hours during balloon use, because bacteria from the cervix or vagina can be introduced into the uterus and the balloon surface can support infection.

Uterotonics can continue when appropriate.

The drainage tube must remain useful.

Flush it every few hours with sterile saline so clots do not block assessment.

If the drainage tube is blocked, the team may be falsely reassured while bleeding collects above the balloon.

The monitoring after placement is not passive.

Monitor vital signs.

Monitor ongoing drainage.

Monitor uterine tone.

Monitor pain, pallor, dizziness, low blood pressure, tachycardia, confusion, abdominal pain, abdominal distension, oliguria, and labs.

The balloon is usually left for about 12 hours, and should not remain more than 24 hours.

Deflation should be gradual, with close monitoring.

If bleeding recurs after emptying, do not simply reinflate or reinsert the balloon.

Recurrent bleeding after removal requires other interventions.

Now the most important Bakri concept: the tamponade test.

The tamponade test asks a simple question.

When the balloon is correctly placed and inflated, does the bleeding stop?

If yes, the test is positive. Continue close monitoring and correction of deficiencies.

If no, the test is negative.

And if the test is negative, Bakri is not a pause button.

Continued heavy bleeding after balloon inflation means tamponade is ineffective. The local protocol says surgery or embolization must be performed.

This is where delay becomes dangerous.

Do not leave a failing Bakri in place because the room feels calmer.

If the blood keeps coming, the balloon has answered the question.

It did not work.

Move.

Now embolization

Embolization closes selected vessels from inside the vascular system.

Interventional radiology performs pelvic angiography, identifies bleeding vessels when possible, and occludes them, commonly with gelatin pledgets or other materials.

The advantages are real.

It can selectively occlude bleeding vessels.

It can preserve the uterus and potential future fertility.

It can sometimes avoid or delay surgery.

It can be useful even in coagulopathy while blood and clotting factor replacement continues.

Gabbe reports cumulative success rates around 90 to 97 percent. The local protocol cites about 80 percent success and about 5 percent complication rate.

Complications include fever, infection, ischemic pain, vascular perforation, tissue necrosis, and thrombosis.

But the key issue is selection.

Embolization is excellent for the right patient and dangerous to wait for in the wrong patient.

The right patient is hemodynamically stable enough for transfer and procedure time.

The bleeding is persistent but slow enough that the clock allows it.

Interventional radiology is available.

The team can coordinate quickly.

The local protocol estimates about 2 hours from decision to completion.

That number should stay in your head.

If the patient is actively collapsing, 2 hours is not a plan.

If the bleeding is slow, persistent, and the patient is stable, embolization may be exactly the right fertility-preserving option.

So the embolization script is:

Stable patient. Slow persistent bleeding. Time and team available. Consider embolization.

And the opposite script is:

Unstable patient. Rapid bleeding. No time for transfer. Do not wait for embolization.

Now surgery

Surgery is not one thing.

Surgery is a family of source-control maneuvers.

Repair bleeding vessels.

Compress the uterus.

Reduce arterial inflow.

Pack for damage control.

Remove the bleeding organ when needed.

At cesarean, the first surgical principles are still direct and local: uterine massage and uterotonics for atony, fluid and blood products, sequential labs, meticulous repair of uterine incision tears, repair of incision angles and extensions, and hemostasis of bleeding vessels.

If these measures do not stop the bleeding, the local protocol moves to uterine artery or internal iliac artery ligation, B Lynch suture with or without balloon integration, and hysterectomy when life-threatening bleeding requires rapid cessation.

Compression sutures work by squeezing the uterus from outside.

The B Lynch suture is the classic name, but the mechanism matters more than the name.

It compresses the uterus so the bleeding surface is pressed closed.

Other compression techniques include Hayman vertical sutures, Pereira sutures, and square sutures.

Compression sutures can be combined with a balloon.

The uterine sandwich technique refers to a B Lynch suture with a Bakri balloon inside the uterine cavity.

Inside pressure plus outside compression.

That is why the image sticks.

The Bakri presses from inside.

The B Lynch compresses from outside.

Uterine artery ligation lowers inflow.

The goal of arterial ligation is to decrease uterine perfusion so bleeding slows enough for hemostasis.

Gabbe describes bilateral uterine artery ligation as a practical initial ligation technique because the uterine artery is accessible.

If bleeding persists, other vessels, such as utero-ovarian or infundibulopelvic vessels, may be addressed when appropriate.

Internal iliac ligation is different.

It can reduce pelvic arterial pressure, but it is technically challenging and time-consuming.

Gabbe cautions that it should not be a first-line technique unless the surgeon is extremely skilled.

That is not a criticism of the operation.

It is a warning about using the wrong tool at the wrong moment with the wrong operator.

In hemorrhage, a technically elegant delay can still be a bad decision.

Now damage-control packing.

Severe prolonged bleeding can push the patient into the vicious cycle of hypothermia, coagulopathy, and metabolic acidosis.

At that point, the operation is not only about anatomy. It is also about physiology.

The local protocol allows packing of the bleeding area, leaving the wound open, transferring the patient to intensive care for monitoring, fluids, blood products, and correction of physiologic disturbances, and then returning to the operating room about 48 hours later for definitive treatment.

Damage control is not giving up.

It is recognizing that the body cannot tolerate a long definitive operation until physiology is restored.

Now hysterectomy

This is the sentence residents must hear clearly:

Hysterectomy is a treatment, not a moral defeat.

The national position paper states that hysterectomy is justified as a primary surgical intervention to save lives or prevent severe complications at the surgeon's discretion.

The local protocol says one should not hesitate to perform hysterectomy in life-threatening bleeding because rapid cessation of bleeding is required to prevent deterioration.

That does not mean hysterectomy is casual.

It means it is legitimate.

It means that when conservative measures are failing and the patient is deteriorating, the ethical obligation is not to preserve the uterus at any cost.

The ethical obligation is to save the patient.

The bravest decision in PPH is sometimes not the most conservative one.

It is the one that stops bleeding while the patient is still salvageable.

The lethal delay is waiting until hypothermia, acidosis, and coagulopathy make every procedure harder.

Let us put the ladder together.

Atony persists after uterotonics.

You inspect the cervix and vagina.

No tear explains the bleeding.

The patient is stable enough for tamponade.

Place Bakri with sterile saline, usually 250 to 300 milliliters, maximum 500 milliliters.

Give antibiotics according to local protocol.

Flush drainage.

Monitor closely.

Use the tamponade test.

If bleeding stops, continue monitoring and resuscitation.

If bleeding continues, do not pretend the balloon is working.

Proceed.

If the patient is stable and bleeding is slow, embolization may be appropriate.

If she is unstable, bleeding rapidly, has suspected rupture, intra-abdominal bleeding, failed tamponade, failed embolization, or no time for embolization, move to laparotomy.

At laparotomy, repair what is bleeding.

Compress the uterus if atony is the problem.

Ligate uterine arteries to reduce inflow.

Use combined techniques when appropriate.

Use packing when damage control is needed.

Perform hysterectomy when life-threatening bleeding requires definitive control.

Now four mechanism lines.

Bakri presses from inside.

B Lynch compresses from outside.

Uterine artery ligation lowers inflow.

Embolization closes selected vessels.

Hysterectomy removes the bleeding organ.

If you know the mechanism, the names become easier.

If you know the clock, the decisions become safer.

Let us close with the resident's escalation script.

We have treated atony medically.

We have excluded a major laceration.

Bleeding continues.

This is no longer a medication-only situation.

Place Bakri if appropriate and use the tamponade test.

If it fails, move to embolization only if she is stable and the bleeding is slow enough.

If she is unstable, we go to surgery.

If conservative surgery fails or delay becomes dangerous, hysterectomy is life-saving treatment.

In the next episode, we will focus on massive transfusion, coagulopathy, and DIC. We will learn why hemoglobin is late, why fibrinogen matters, how blood product strategy changes in massive obstetric bleeding, and how to protect the patient from the lethal triad of hypothermia, acidosis, and coagulopathy.